Lab no. 4

Simple Python Database Connections

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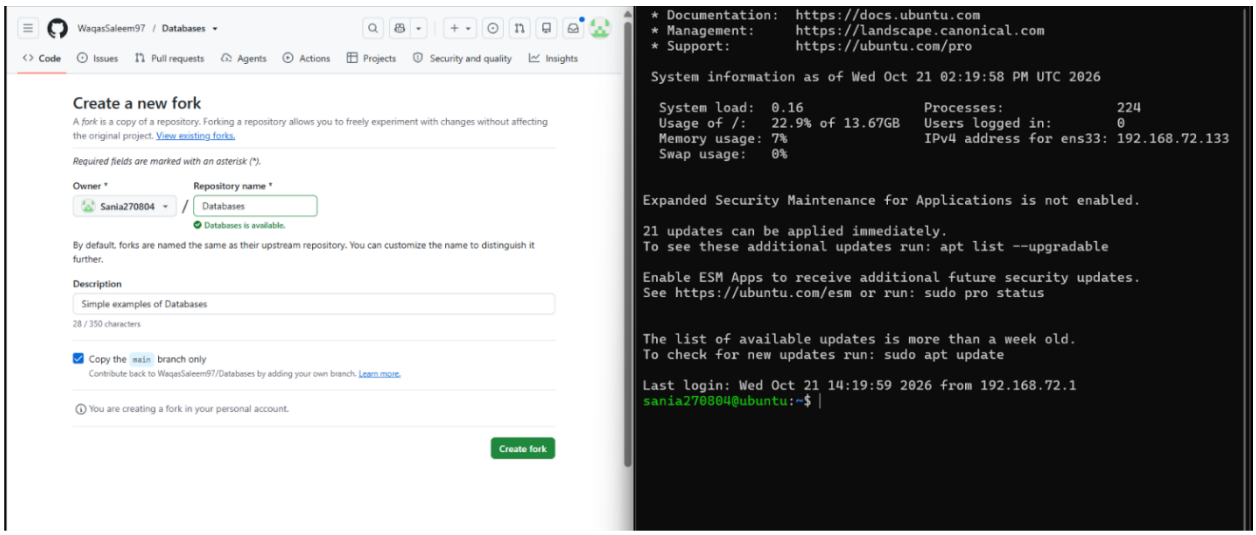
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COURSE CODE: CS-649

SECTION: A

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Task 1: Fork and Clone the Databases Repository



Open Databases, select Fork, and keep the name Databases.

```
sania270804@ubuntu:~$ cd ~
sania270804@ubuntu:~$ git clone git@github.com:Sania270804/Databases.git
Cloning into 'Databases'...
remote: Enumerating objects: 101, done.
remote: Counting objects: 100% (101/101), done.
remote: Compressing objects: 100% (90/90), done.
remote: Total 101 (delta 30), reused 1 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (101/101), 39.86 KiB | 141.00 KiB/s, done.
Resolving deltas: 100% (30/30), done.
sania270804@ubuntu:~$ cd Databases
sania270804@ubuntu:~/Databases$
```

Clone your own fork.

Task 2: Install Docker, Docker Compose, and Python on Ubuntu

```
sania270804@ubuntu:~$ git --version
git version 2.53.0
sania270804@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
sania270804@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
sania270804@ubuntu:~$
```

Verify the Git, Docker, and Docker Compose installations.

Task 3: Start the Database Services

```
(.venv) sania270804@ubuntu:~/Databases$ docker compose up -d --wait mysql mongo-dev mongo-test mongo-prod
[+] up 29/33
 ✔Image mongo:8.0                 Pulled                                                                    289.8s
 ✔Image mysql:8.4                 Pulled                                                                    276.1s
 ✔Network databases_default       Created                                                                     0.1s
 ✔Volume databases_mysql-data     Created                                                                     0.0s
 ✔Volume databases_mongo-prod-data Created                                                                    0.0s
 ✔Volume databases_mongo-dev-data  Created                                                                    0.0s
 ✔Volume databases_mongo-test-data Created                                                                    0.0s
 ⠴ Container databases-mysql-1     Waiting                                                                    3.7s
 ⠴ Container databases-mongo-test-1 Waiting                                                                   3.7s
 ⠴ Container databases-mongo-prod-1 Waiting                                                                   3.7s
 ⠴ Container databases-mongo-dev-1  Waiting                                                                   3.7s
container databases-mongo-prod-1 exited (1)
(.venv) sania270804@ubuntu:~/Databases$ docker compose ps
NAME               IMAGE       COMMAND                  SERVICE   CREATED          STATUS                          PORTS
databases-mysql-1   mysql:8.4   "docker-entrypoint.s…"   mysql     10 seconds ago   Up 8 seconds (health: starting)   33060/tcp, 127.0.0.1:3307->3306/tcp
(.venv) sania270804@ubuntu:~/Databases$
```

Start the four supplied databases.

```
(.venv) sania270804@ubuntu:~/Databases$ docker compose port mysql 3306
127.0.0.1:3307
(.venv) sania270804@ubuntu:~/Databases$ docker compose port mongo-dev 27017
127.0.0.1:27017
(.venv) sania270804@ubuntu:~/Databases$ docker compose port mongo-test 27017
127.0.0.1:27018
(.venv) sania270804@ubuntu:~/Databases$ docker compose port mongo-prod 27017
127.0.0.1:27019
(.venv) sania270804@ubuntu:~/Databases$
```

Inspect the ports used by Python on the Ubuntu host.

Task 4: MySQL with Values in the Code

```
(.venv) sania270804@ubuntu:~/Databases$ nano db_connect.py
(.venv) sania270804@ubuntu:~/Databases$ python db_connect.py
Connected to MySQL: school_mysql
Server: 127.0.0.1 3307
Environment: dev
(.venv) sania270804@ubuntu:~/Databases$
```

Run the file.

Task 5: MongoDB with Values in the Code

```
(.venv) sania270804@ubuntu:~/Databases$ nano db_connect.py
(.venv) sania270804@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) sania270804@ubuntu:~/Databases$
```

MongoClient(uri) creates the MongoDB client, and client[database_name] selects the database.

username, password, endpoint and database_name hold the settings separately. The + operator joins them into uri.

Task 6: MongoDB with os.getenv

```
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) sania270804@ubuntu:~/Databases$
```

Set the dev values in the same Ubuntu shell and run.

```
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) sania270804@ubuntu:~/Databases$
```

Set the test values and run.

```
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) sania270804@ubuntu:~/Databases$
```

Set the simulated prod values and run.

Task 7: MongoDB with a Credentials File

```
(.venv) sania270804@ubuntu:~/Databases$ cp -n credentials.ini.example credentials.ini
ls credentials.ini
cp: warning: behavior of -n is non-portable and may change in future; use --update=none instead
credentials.ini
```

Create one local file containing the three environment sections.

```
(.venv) sania270804@ubuntu:~/Databases$ chmod 600 credentials.ini
stat -c '%a %n' credentials.ini
600 credentials.ini
(.venv) sania270804@ubuntu:~/Databases$
```

Restrict the file to your account and check its mode

```
(.venv) sania270804@ubuntu:~/Databases$ export APP_ENV=dev
python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) sania270804@ubuntu:~/Databases$
```

Choose dev and run.

```
(.venv) sania270804@ubuntu:~/Databases$ export APP_ENV=test
python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) sania270804@ubuntu:~/Databases$
```

Choose test and run.

```
(.venv) sania270804@ubuntu:~/Databases$ export APP_ENV=prod
python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) sania270804@ubuntu:~/Databases$
```

Choose simulated prod and run.

Link of forked Databases Repository:

https://github.com/Sania270804/Databases